Category	Solved 1 row (teleoperated)	Partially solved 22 rows		Unsolved 17 rows		Contested 0 rows
Perception 8 rows		◇ P02† Is biplane needed? (D5)	■ P03 Tip tracking on clinical X-ray	◆ P01 Single-view tip depth and roll sign	■ P07 Dose-aware imaging on demand	None: no debate (D1–D6) met the plan §1 definition
		■ P04† Camera bench versus real X-ray	■ P05 Live C-arm frames and latency			
		◇ P06 2D/3D registration under motion	◇ P37 Clot detection from proximal force			
Modelling 9 rows		■ P08 Static device-shape prediction	■ P12 Transmission error, roller slip (H5)	○ P09 Predicting tip and roll over time	◆ P10 Wall friction from proximal Fz/Tz	
		○ P14† Sim-to-real, rigid phantoms (D4)	○ P16 Open-loop navigation by model (D3)	◇ P11 Torsional wind-up and whip	■ P13 Wall deformation, flow, motion in simulators	
				○ P15 Sim-to-real gap in tissue		
Control 4 rows		■ P17 Closed loop versus model accuracy (H1)	◇ P23† Arch-to-LCCA with a straight-tip catheter	■ P21 Residual policy on base controller (H3)		
		◇ P25 Hard force limit during autonomy				
Learning 5 rows		■ P19† Modular versus end-to-end (D2)	■ P22† Unseen anatomy under fluoroscopy	■ P18† Ranking IL, RL and hybrid (D1)	■ P20† World model versus model-free RL	
		◇ P24† Two coordinated devices (H8)				
System 7 rows	■ P38 Cerebral angiography in humans *teleoperation, not autonomy*	■ P26*† Failure detection and handover	■ P27† Same tracker closes and scores (H7)	◇ P28 Independent safety supervisor	■ P32 Bench logs as training data (H6)	
		◇ P29 Proximal Fz/Tz event sensing	■ P31 Same policy, SOFA twin and robot			
Translation 6 rows		■ P34 Real-X-ray autonomy beyond bench	■ P40† Replication by an independent group	■ P30* No measured perforation force	■ P33† Neurovascular autonomy in vivo (D6)	
				◆ P35 Swine lack a circle-of-Willis route	◆ P36 Ex vivo lacks flow and clotting	
Regulation 1 row				■ P39 Human autonomy and regulatory path		

■ Engineering (23) ◇ Mixed: engineering and intrinsic parts (9) ◆ Intrinsic (4) ○ Unknown so far (4)

solved partially solved unsolved label states insufficient evidence (7)

* row relies on a card with verified = false (2) † row relies on fluoroscopy counts scored by the EC-0447 tip tracker (13) P## = row ID in the report's matrix table